Projeto

Título
Spec-Driven for Agentic Software Development: A Characterisation and Comparison of Existing Approaches

Problema
AI coding agents now carry out complete development tasks: they analyse repositories, plan changes, modify multiple files and run tests. Spec-Driven Development (SDD) proposes that the specification should be the central artefact of this work, giving the agent structured context (Böckeler, 2025; Piskala, 2026).

Since 2025, several approaches have emerged that implement SDD in different ways, such as GitHub Spec Kit, OpenSpec and the BMAD Method (Macedo, 2026), and Kiro (Böckeler, 2025). They differ in the artefacts they require, the phases of the process and the points of human review (Macedo, 2026).

The evidence base on SDD is still immature and relies mainly on grey literature (Díaz et al., 2026), so a team that wants to adopt SDD has no comparable evidence on which to base its choice. The available comparisons are mostly grey literature (e.g., Böckeler, 2025). The academic works identified include conceptual analyses (Díaz et al., 2026; Piskala, 2026), a process taxonomy (Macedo, 2026) and a small number of empirical studies, on SDD or on related context mechanisms (Gloaguen et al., 2026; Panda, 2026). The only controlled comparison of SDD frameworks identified in the preliminary literature search compares three of them on output determinism and hallucination detection, without measuring cost or human effort (Panda, 2026).

A process taxonomy of SDD frameworks already exists, but it points out the lack of end-to-end benchmarks (Macedo, 2026). The problem to be addressed is the lack of a comparison of the results of the existing SDD standards when applied to the same project, supported by an up-to-date common characterisation of them.

1. Envolve questões técnicas amplas ou de difícil harmonização?
Sim

Se sim, justificação
The SDD approaches use processes, artefacts and tools that are not compatible with each other. Comparing them requires a common framework and an evaluation protocol that separates the effect of the approach from the effect of the agent, the model and the operator.

2. Tem uma solução óbvia?
Não

Se não, justificação
Evidence on SDD is scarce (Díaz et al., 2026) and points in different directions. Proponents argue that SDD adds value in certain contexts (Piskala, 2026), while a controlled study on repository context files, a related context mechanism rather than SDD itself, found higher inference cost without an overall gain in task success (Gloaguen et al., 2026). It is not known which approach pays off in which situation.

3. Aborda problemas não abrangidos pelas normas e códigos atuais?
Sim

Se sim, justificação
There is no formal standard for specifications written for coding agents. The existing SDD

approaches are de facto conventions defined by vendors or communities, which are still converging on common practices (Macedo, 2026).

4. Envolve diversos grupos de partes interessadas (stakeholders)?
Não

Se sim, justificação

5. Inclui muitas partes componentes ou subproblemas, envolvendo múltiplas disciplinas?
Sim

Se sim, justificação
The problem combines requirements engineering, software architecture, LLM-based coding agents, software quality and empirical research methods. It breaks down into identifying the standards, designing a reference project, instrumenting the experimental runs, analysing the results and building a decision guide.

6. Tem consequências significativas numa variedade de contextos?
Não

Se sim, justificação

Objetivos
Main objective: to identify and characterise the existing Spec-Driven Development standards for development with agents, and to compare their effect on the total cost and the quality of development when applied to the same project.

Specific objectives:
1. Identify the existing SDD standards and characterise them in a common comparative framework: what each one prescribes and how it can be used (Macedo, 2026).
2. Design a reproducible evaluation protocol, with a reference project and cost and quality metrics defined in advance.
3. Apply each approach to the same example project, together with a baseline without a structured specification, adapting from Gloaguen et al. (2026) the comparison of each condition with a no-context baseline on the same tasks; that study examines context files, not SDD, so only this design element is reused.
4. Compare the results of the approaches, with each other and with the baseline.
5. Produce a decision guide on which approach to use in which situation.

Research question: Which Spec-Driven Development approaches exist for development with agents, how do they differ, and how do they affect the total cost and the quality of development when applied to the same project?

Haverá interpretação do problema a resolver?
Sim

Se sim, como?
The causes of the problem (the recent proliferation of SDD approaches without comparative evaluation), the stakeholders and their interests, and the consequences of choosing an approach without evidence are analysed. The problem is examined from several perspectives (cost, quality, human effort and platform dependence), confronting the benefits expected by proponents of SDD (Piskala, 2026) with indirect evidence from related context mechanisms (Gloaguen et al., 2026).

Haverá sistematização do conhecimento existente para a resolução do problema?
Sim

Se sim, como?
A structured state of the art with a documented protocol: research questions, search strings, inclusion and exclusion criteria, and a PRISMA flow of the selection. The main sources are Scopus, Web of Science, IEEE Xplore and the ACM Digital Library. Tool documentation and repositories are included as primary sources for the approaches themselves, assessed with explicit quality criteria. The result is the identification of the existing SDD standards and a comparative framework describing what each one prescribes and how it can be used (Böckeler, 2025; Macedo, 2026).

Haverá avaliação de diferentes abordagens para a resolução do problema?
Sim

Se sim, como?
At two levels. In the state of the art, existing ways of evaluating SDD and coding agents (benchmarks, controlled experiments [Gloaguen et al., 2026; Panda, 2026] and industrial comparisons) are compared to justify the design of the evaluation. In the experiment, a set of SDD approaches selected with explicit criteria is applied to the same reference project and compared with each other and with a baseline without a structured specification.

Haverá conceção de uma solução (produto, componente, processo)?
Sim

Se sim, como?
A reproducible evaluation protocol is designed, including the reference project, a neutral requirements brief, acceptance tests, the experimental conditions, cost and quality metrics defined in advance, and rules for human intervention. A decision guide is designed that relates SDD approach, type of task and context.

Haverá construção de uma solução?
Sim

Se sim, como?
The reference project is implemented with each selected approach and with the baseline, using the coding agents each approach supports. The execution environment and scripts for automatic metric collection are built, and the artefacts are kept in a repository to allow replication. The decision guide is produced from the results.

Haverá avaliação da solução desenhada/implementada?
Sim

Se sim, como?
The protocol is evaluated by its repeatability, defined as the variation of the cost and quality metrics between repeated runs of the same condition, a concern motivated by the output non-determinism reported by Panda (2026), and by its ability to distinguish the approaches. The decision guide is evaluated by applying it to tasks not used to build it and checking whether its recommendation matches the observed results. Threats to validity are discussed.

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